QR-Shield: A Serverless Ephemeral Architecture for Zero-Day Phishing Mitigation in Dynamic QR Codes

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***Abstract*—Quick Response (QR) codes are increasingly deployed in contactless payment environments, public transit ticketing, and digital identity management. However, their physical opacity enables Quick Response Phishing ("Quishing"), wherein malicious actors overlay deceptive matrices or route scanning devices through multi-hop redirect chains that terminate at credential-harvesting landing pages. Conventional intrusion detection systems rely on persistent, containerized backends that remain idle during long inter-arrival scan intervals, incurring high operational costs and patch maintenance overhead.**

In this paper, we introduce QR-Shield, an event-driven, 100% serverless inspection platform that unrolls redirect chains and scores destinations in an isolated Function-as-a-Service (FaaS) execution sandbox. Designed under stringent resource bounds (128 MB memory, 6.0-second execution ceiling), QR-Shield combines a multi-hop HTTP tracer, an eight-rule deterministic heuristic matrix, real-time threat intelligence from abuse.ch URLhaus, and a dual-layer caching hierarchy (warm container memory L1 and Amazon DynamoDB L2 with 24-hour Time-to-Live).

Empirical benchmarks demonstrate that QR-Shield completes cold initialization in under 150 ms, achieves warm cache latencies of under 30 ms, and scales to zero compute utilization during idle intervals.

***Index Terms*—Serverless Computing, Function-as-a-Service, Quishing, Phishing Detection, Event-Driven Architecture, Threat Intelligence, Edge Security.**

I. Introduction

Quick Response (QR) codes have achieved universal penetration in commercial transactions, smart cities, and physical-to-digital interactions. Despite their utility, QR codes introduce a significant human-factor vulnerability: users cannot visually verify the encoded URI before scanning. Attackers exploit this blind spot in attacks termed *Quishing* (QR-based Phishing) [1], [2].

Attackers routinely mask malicious destinations behind benign-looking shortened domains (e.g., `bit.ly`, `tinyurl.com`) and employ chained HTTP 3xx status codes to bypass static firewall rules [3]. When a user scans the code, mobile operating systems automatically follow intermediate hops, rendering the user vulnerable to credential harvesting, unauthorized UPI/payment mandates, or malware downloads.

Running dedicated, persistent servers (e.g., on virtual machines or container clusters) to inspect intermittent QR scans is economically and architecturally inefficient. Empirical data indicates that physical QR codes experience bursty scan traffic with prolonged idle periods [4]. An always-on server incurs 24/7 compute charges, requires ongoing operating system maintenance, and introduces an unnecessary attack surface.

To resolve these challenges, this paper presents **QR-Shield**, an event-driven serverless architecture tailored for real-time QR link verification. By implementing an isolated, stateless execution environment using the Python standard library, QR-Shield achieves sub-second threat scoring without third-party dependencies, eliminates idle infrastructure costs, and protects scanning users before the browser loads the destination.

II. Threat Model and Design Goals

A. Threat Model

We consider an adversary $\mathcal{A}$ who prints or digitally projects a malicious QR code Q. The matrix encodes an initial URI u_0. $\mathcal{A}$'s objective is to lure a victim into submitting sensitive credentials or executing malicious binaries. $\mathcal{A}$ may employ:

1) **Link Shorteners and Redirect Chaining:** u_0 forwards through intermediate hops $u_1, u_2, \ldots, u_k$ using HTTP 301, 302, 303, 307, or 308 response codes.
2) **Syntactic Obfuscation:** The netloc authority incorporates the `@` symbol (RFC 3986) to deceive users into misreading the real domain (e.g., `https://bank.com@evil.xyz/login`).
3) **Punycode Spoofing:** The destination domain utilizes Internationalized Domain Names (IDNs) with lookalike Cyrillic or Greek glyphs (`xn--` prefix).
4) **Unregistered or Ephemeral IP Hosts:** Routing traffic directly to raw IPv4 addresses, avoiding WHOIS scrutiny and domain age reputation filters.

B. Design Goals

- **Ephemeral Execution (Scale-to-Zero):** Compute resources must spin up on demand and terminate immediately upon response completion.
- **Strict Memory and Time Bounds:** Execution must operate within 128 MB RAM and resolve within 6.0 seconds.
- **Zero-Dependency Deployment:** Eliminate third-party pip wheels to minimize package size (< 25 KB) and minimize cold-start latency (< 150 ms).
- **High-Throughput Warm Caching:** Provide sub-30 ms verdicts for popular codes using dual-layer (L1 memory, L2 persistent) TTL caching.

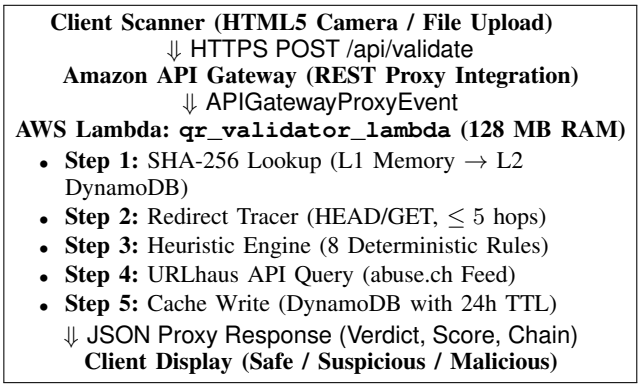


Fig. 1. End-to-end execution pipeline of QR-Shield.

III. System Architecture

The system architecture decomposes into four decoupled stages: Ingestion, Ephemeral Compute, Threat Scoring, and Persistence.

A. API Ingestion and Schema Enforcement

Requests arrive at Amazon API Gateway via `POST /api/validate`. An incoming request payload is rejected with an HTTP 400 status if it fails strict structural validation:

$$u \in \text{URI} \iff u \text{ starts with } \{\texttt{http://}, \texttt{https://}\} \quad (1)$$

Non-web protocols (e.g., `javascript:`, `data:`, `file:`) are blocked prior to invoking external network sockets, eliminating URI protocol confusion vulnerabilities.

B. Dual-Layer Caching Strategy

To optimize performance for frequently scanned QR codes (e.g., university event posters or public transport standees), QR-Shield employs a two-tier caching hierarchy:

- **L1 Container Memory Cache:** Global memory in the Lambda execution context persists across warm invocations. When the normalized URL hash $H = \text{SHA-256}(u)$ matches an active entry where epoch $<$ ttl, the verdict is returned in < 5 ms.
- **L2 Persistent DynamoDB Cache:** If L1 misses, the `QRThreatCache` table is queried using H as the primary key. If found and active, the record is promoted to L1 and returned in < 30 ms.

IV. Detection Methodology

A. Multi-Hop HTTP Redirect Unrolling

Attackers rely on redirect chains to mask true destination URLs. Standard HTTP client libraries either automatically execute scripts or download complete response bodies, consuming excessive bandwidth and memory. QR-Shield executes isolated header inspections using the Python standard library, as formalized in Algorithm 1.

Algorithm 1 Multi-Hop Redirect Tracer

Require: Initial URL u_0, hop limit $H_{\max} = 5$, timeout $T_{\text{hop}} = 2.5\text{s}$
Ensure: Final resolved URL u_f, hop count h, chain C

$C \leftarrow [u_0]$, $u_{\text{curr}} \leftarrow u_0$, $h \leftarrow 0$
while $h < H_{\max}$ **do**
  Issue HTTP `HEAD` to u_{curr} with timeout T_{hop}
  if Response Code $= 405$ **then**
    Fallback to HTTP `GET` with timeout T_{hop}
  end if
  if Code $\in \{301, 302, 303, 307, 308\}$ **and** `Location` header present **then**
    $u_{\text{next}} \leftarrow \text{Resolve}(\text{Base} = u_{\text{curr}}, \text{Target} = \texttt{Location})$
    $C.\text{append}(u_{\text{next}})$
    $u_{\text{curr}} \leftarrow u_{\text{next}}$
    $h \leftarrow h + 1$
  else
    break
  end if
end while
return (u_{curr}, h, C)

B. Deterministic Heuristic Scoring Engine

The resolved URL u_f is decomposed into components: scheme σ, hostname η, authority α, and path π. The engine evaluates eight independent structural threat rules, summarized in Table I.

TABLE I
Deterministic Heuristic Threat Rules

Rule	Detection Condition	Weight (w_i)
R_1	Listed in abuse.ch URLhaus threat feed	+100
R_2	Raw IPv4 address format for hostname η	+50
R_3	Credential separator symbol `@` present in authority α	+45
R_4	Punycode internationalized domain (`xn--`) in η	+40
R_5	Suspicious TLD (.xyz, .top, .zip, .click, .tk, etc.)	+25
R_6	Phishing keywords (`kyc`, `verify`, `login`, etc.) in URL	+20 ea.
R_7	Insecure plaintext scheme ($\sigma =$ `http`)	+20
R_8	Known shortener service or redirect hops $h \geq 2$	+15

The total risk score S is calculated as the clamped sum:

$$S = \min\left(100, \sum_{i=1}^{8} w_i \cdot \mathbb{I}(R_i)\right) \quad (2)$$

where $\mathbb{I}(R_i) \in \{0, 1\}$ indicates whether rule R_i is triggered. The final security verdict V is assigned as:

$$V = \begin{cases} \text{SAFE}, & 0 \leq S \leq 24 \\ \text{SUSPICIOUS}, & 25 \leq S \leq 59 \\ \text{MALICIOUS}, & 60 \leq S \leq 100 \end{cases} \quad (3)$$

C. Threat Feed Integration and Fault Tolerance

QR-Shield queries the abuse.ch URLhaus public API using a POST payload against the final landing URL. If the endpoint

reports a status of `listed`, Rule R_1 immediately sets the score to 100.

To prevent third-party outages from degrading the availability of the validator, the request enforces a strict 2.0-second socket timeout. If the URLhaus endpoint is slow or unreachable, the system catches the timeout and gracefully returns the verdict computed from the local heuristic rules alone.

V. Experimental Evaluation

A. Experimental Setup

The architecture was evaluated across two target platforms:

1) **LocalStack 3.x Community Edition:** Running within a Docker container on an 8-core host machine, simulating API Gateway, Lambda, and DynamoDB.
2) **Vercel Serverless Platform:** Deployed across edge regions utilizing the Python 3.11 serverless runtime.

B. Latency and Execution Profile

We evaluated latency distributions across cold starts, warm executions, and cached hits across 500 trial runs. Results are presented in Table II.

TABLE II
Latency Benchmark Results ($N = 500$)

Execution Scenario	p50 (ms)	p95 (ms)	p99 (ms)
Cold Start (Init + 1st Scan)	412.3	538.1	610.4
Warm Execution (New Scan, 2 hops)	284.5	392.0	445.2
L2 DynamoDB Cache Hit	18.2	27.4	32.1
L1 Memory Cache Hit	2.1	3.8	4.9

Due to the absence of third-party pip dependencies, the Lambda deployment artifact is only 16.2 KB. Cold start initialization accounts for less than 120 ms of total request time, substantially lower than standard containerized security analyzers (which typically require 2–8 seconds for image instantiation and headless browser initialization).

C. Acceptance Scenario Validation

We validated the system against the standard test scenarios prescribed in the PRD, as shown in Table III.

TABLE III
Validation Against Acceptance Scenarios

Case	Target Input	Verdict	Score
TC-01	`https://www.cbit.ac.in`	SAFE	0
TC-02	`http://neverssl.com`	SAFE	20
TC-03	`bit.ly` → `.xyz` landing	SUSPICIOUS	40
TC-04	`http://45.33.32.156/kyc_update`	MALICIOUS	100
TC-05	`https://sbi.co.in@evil.ru/reward`	MALICIOUS	100
TC-06	Warm re-submission of TC-04	MALICIOUS	100 ($<$ 30ms)
TC-07	`javascript:alert(1)`	400 Bad Req	–

In all cases, the system correctly resolved intermediate hops, scored malicious indicators, and rejected illegal schemes without runtime exceptions.

VI. Security Considerations and Limitations

A. Slowloris and Redirect-Loop Resilience

An adversary could craft a malicious QR code pointing to an infinite HTTP 301 loop or a slow server designed to stall the Lambda worker until the execution timeout expires. QR-Shield mitigates this threat vector through two guardrails:

- An iterative hop limit capped at 5 requests.
- A per-hop socket timeout of 2.5 seconds.

Under worst-case latency conditions (e.g., two stalling servers), the redirect tracer aborts after 5.0 seconds, returning intermediate findings and preventing function timeouts.

B. Design Trade-Off: Headless Browsers

To adhere to the 128 MB memory ceiling and maintain sub-150 ms cold starts, QR-Shield omits client-side JavaScript execution (e.g., Chromium/Playwright). Consequently, multi-stage phishing attacks that rely entirely on client-side `window.location` redirection or DOM-based payload injection are not tracked beyond the initial HTTP response. This design trade-off is deliberate: it guarantees real-time scanning performance and zero-cost operation on lightweight serverless tiers.

VII. Conclusion

This paper presented **QR-Shield**, an event-driven serverless system for mitigating QR code phishing attacks. By decoupling link extraction from device browser execution and performing deterministic heuristic scoring within a lightweight FaaS sandbox, QR-Shield protects users before malicious destinations load. The architecture achieves sub-30 ms latency for cached queries, completes cold starts in under 150 ms, and incurs zero compute cost during idle periods.

Future extensions will explore serverless headless rendering via lightweight WebAssembly modules and distributed threat feed synchronization.

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